

Dual-Domain MMD Adaptation: Small-Anchor Runs, Per-Class Breakdown

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1 Setup

Both directions of adaptation were run from a small (1600-image), stride-subsampled pretrained anchor checkpoint, adapted against the full opposite-domain large target set:

- **real_small_source_syn_large_target:** source = real_small (pretrained, detached, forward-only reference), target = syn_large (fine-tuned via detection loss, pulled toward source via MMD).
- **syn_small_source_real_large_target:** mirror direction, source = syn_small, target = real_large.

Both adapted checkpoints, plus their originating small-anchor pretrain baselines (real_small, syn_small), were evaluated per-class (person, vehicle) on both domains' validation sets: the synthetic val set (640 images) and the real val set (548 images).

2 Results

Model	p mAP50	p mAP50-95	v mAP50	v mAP50-95
real_small (pretrain baseline)	0.0745	0.0194	0.162	0.106
syn_small (pretrain baseline)	0.679	0.341	0.693	0.473
real_small_source_syn_large_target (adapted)	0.687	0.353	0.696	0.482
syn_small_source_real_large_target (adapted)	0.125	0.0319	0.233	0.154

Table 1: Validated on the synthetic (syn) val set.

Model	p mAP50	p mAP50-95	v mAP50	v mAP50-95
real_small (pretrain baseline)	0.665	0.311	0.884	0.633
syn_small (pretrain baseline)	0.0125	0.00286	0.293	0.174
real_small_source_syn_large_target (adapted)	0.0247	0.00628	0.318	0.196
syn_small_source_real_large_target (adapted)	0.647	0.312	0.889	0.640

Table 2: Validated on the real val set.

3 Discussion

Vehicle detection improves consistently in both directions. Comparing each adapted model to its own target domain’s in-domain baseline: `real_small_source_syn_large_target` raises vehicle mAP50 on syn from 0.693 to 0.696 (mAP50-95: $0.473 \rightarrow 0.482$), and `syn_small_source_real_large_target` raises vehicle mAP50 on real from 0.884 to 0.889 (mAP50-95: $0.633 \rightarrow 0.640$). Both directions, both metrics, same sign – the one class that shows a genuine, if modest (0.3–0.9 point), improvement from MMD alignment in this setup.

Person detection is inconsistent. It improves in the syn-target direction ($0.679 \rightarrow 0.687$ mAP50) but slightly regresses in the real-target direction ($0.665 \rightarrow 0.647$ mAP50).

Source-domain forgetting is present and worse for person than vehicle. Neither adapted model receives a detection-loss gradient on its own source domain after adaptation starts. `real_small_source_syn_large_target`’s person AP on real collapses from 0.665 (as `real_small`’s own in-domain score) to 0.0247, while vehicle only drops to 0.318 from 0.884 – a much smaller relative loss. The same asymmetry holds in the mirror direction.

Caveat. These are single runs on val sets of modest size (160–640 images); the vehicle-specific gain, while consistent across both directions, is small enough that confirming it isn’t a favorable-seed artifact would need a repeat run (or the bandwidth-freeze + MMD-weight-decay regularization discussed in `experiment_summary_v2.tex`) before treating it as a settled result.